

RESEARCH ARTICLE

Deep learning-based pigment analysis model trained with optical approach and ground truth assistance

Geunho Jung  | Semin Kim | Jongha Lee | Sangwook Yoo

AI R&D Center, Lululab Inc., Seoul,
Republic of Korea

Correspondence

Geunho Jung, AI R&D Center, Lululab
Inc., Seoul, Republic of Korea.
Email: gh.jung@icloud.com

Funding information

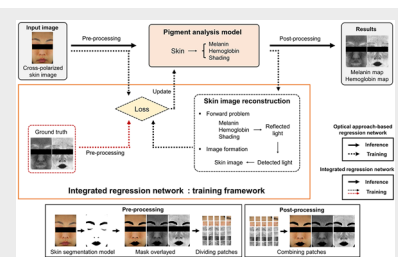
NIPA

Abstract

This study introduces an integrated training method combining the optical approach with ground truth for skin pigment analysis. Deep learning is increasingly applied to skin pigment analysis, primarily melanin and hemoglobin. While regression analysis is a widely used training method to predict ground truth-like outputs, the input image resolution is restricted by computational resources. The optical approach-based regression method can alleviate this problem, but compromises performance. We propose a strategy to overcome the limitation of image resolution while preserving performance by incorporating ground truth within the optical approach-based learning structure. The proposed model decomposes skin images into melanin, hemoglobin, and shading maps, reconstructing them by solving the forward problem with reference to the ground truth for pigments. Evaluation against the VISIA system, a professional diagnostic equipment, yields correlation coefficients of 0.978 for melanin and 0.975 for hemoglobin. Furthermore, our model can produce pigment-modified images for applications like simulating treatment effects.

KEYWORDS

deep learning, diffuse Optics, skin analysis, skin pigment



1 | INTRODUCTION

The coloration of human skin is influenced by the presence of melanin and hemoglobin, which are prominent chromophores in the visible wavelength range. Melanin absorbs light across the entire spectrum with an exponential decrease from blue to red, contributing to the observed darkness and yellowish color [1]. Conversely, hemoglobin significantly absorbs green light, leading to the generation of a complementary red color [2]. Quantifying melanin and hemoglobin concentrations through distribution maps offers an accurate method to assess dermatological conditions, such as erythema and hyperpigmentation [3].

Spectral reflectance techniques are commonly used to measure skin melanin and hemoglobin concentrations [4]. Clinical devices either analyze the reflectance ratios at different wavelengths to determine melanin and hemoglobin indices [5, 6] or infer their concentrations from the absorption spectrum through methods like diffuse reflectance spectroscopy (DRS) [7]. These chromophore analysis techniques can be applied to image pixels, expanding the analysis from point measurement to spatial distribution. Whole optical-based image processing approaches are prevalent in commercial equipment and efficient enough for clinical use [8, 9], but most analysis tools are restricted to the manufacturer's imaging equipment, limiting versatility. The independent component

analysis (ICA), which is one of the representative methods, requires significant pigment concentration variation within the image to distinguish color components [10]. Therefore, the process may involve manually sliding the window to identify analyzable regions within the image. Additionally, image analysis requires substantial computational time due to repetitive calculations such as least-square fitting [11] or ICA performed on all pixels.

Deep learning-based approaches have recently gained popularity in skin pigment analysis, especially in the segmentation of erythema and hyperpigmented areas [12–14]. Regression analysis is a commonly used technique that a neural network architecture learns the relationship between input images and ground truths to predict desired outputs. However, this method requires the preparation of ground truth corresponding to the training dataset and faces limitations when applied to high-resolution images due to high memory requirements, being computationally expensive, and the relatively small receptive field of convolutional filters compared to the image resolution. To alleviate these restrictions, our previous research applied the optical approach to train a pigment analysis model, providing an alternative to preparing ground truth for regression analysis [15]. This model decomposes skin images into melanin, hemoglobin, specular, and shading maps, after which the forward problem is solved to create skin images. For clarity in terminology, we use the term “reconstructed image” for the theoretically created skin image based on the provided pigments and geometric conditions (represented as output maps), while the full-sized skin image employed for analysis is denoted to as the “input image” throughout this study. The model is trained to minimize the discrepancy between the input image and the reconstructed image, with the ultimate goal of generating accurate pigment maps that resemble actual skin conditions. Although regression analysis methods that use optical theory to substitute for ground truth achieve reasonable results and can analyze images at a higher resolution than the model's input size, their performance is not as good as when directly using ground truth.

In this study, we propose an integrated training method that combines the optical approach with ground truth for pigment analysis. This method adopts the learning structure of the optical approach while incorporating ground truth for pigments to simplify the learning process. We evaluate the performance of our integrated training method in comparison to regression analysis methods: one using ground truth and the other employing the optical approach to substitute for preparing ground truth. By leveraging the strengths of separate approaches, the integrated training method can analyze

higher resolution images beyond the input size limitation, achieve performance comparable to regression analysis, and generate pigment-modified images for various applications.

2 | MATERIALS AND METHODS

The training and inference frameworks for the pigment analysis model is described in Figure 1. To clearly distinguish among the training frameworks for regression analysis, we use the term “simple regression network” for the model trained using prepared ground truth, “optical approach-based regression network” for the model utilizing the optical approach as a substitute for preparing ground truth, and “integrated regression network” for the model that combines the optical approach with ground truth, incorporating ground truth within the optical approach-based learning structure. Similarly, the training methods for each network are sequentially referred to as “simple regression method,” “optical approach-based regression method,” and “integrated regression method.” These terms will be consistently used in the subsequent sections.

2.1 | Model architecture

Our model is based on the ResUNet++ architecture (Figure 2), which combines residual blocks, squeeze and excitation blocks, astrous spatial pyramid pooling (ASPP), and attention blocks, all integrated into the UNet structure. Previous studies have demonstrated that the ResUNet++ architecture outperforms both UNet and ResUNet architectures [16].

To optimize memory utilization, we have adapted the architecture by halving the number of channels. The modified version serves as the network architecture for the pigment analysis model across all three training methods, as well as for the skin segmentation model.

2.2 | Optical approach

In our previous research, we utilized skin decomposition principles from computer vision [17] to train a pigment analysis model [15]. This model, based on a UNet structure, decomposes skin regions into four components: melanin, hemoglobin, shading, and specular maps. Although the model yields reasonable analysis results, it faces training challenges from extensive exploration of output combinations and the significant time required for the training process stabilization. Under identical training conditions and with the same dataset, there have

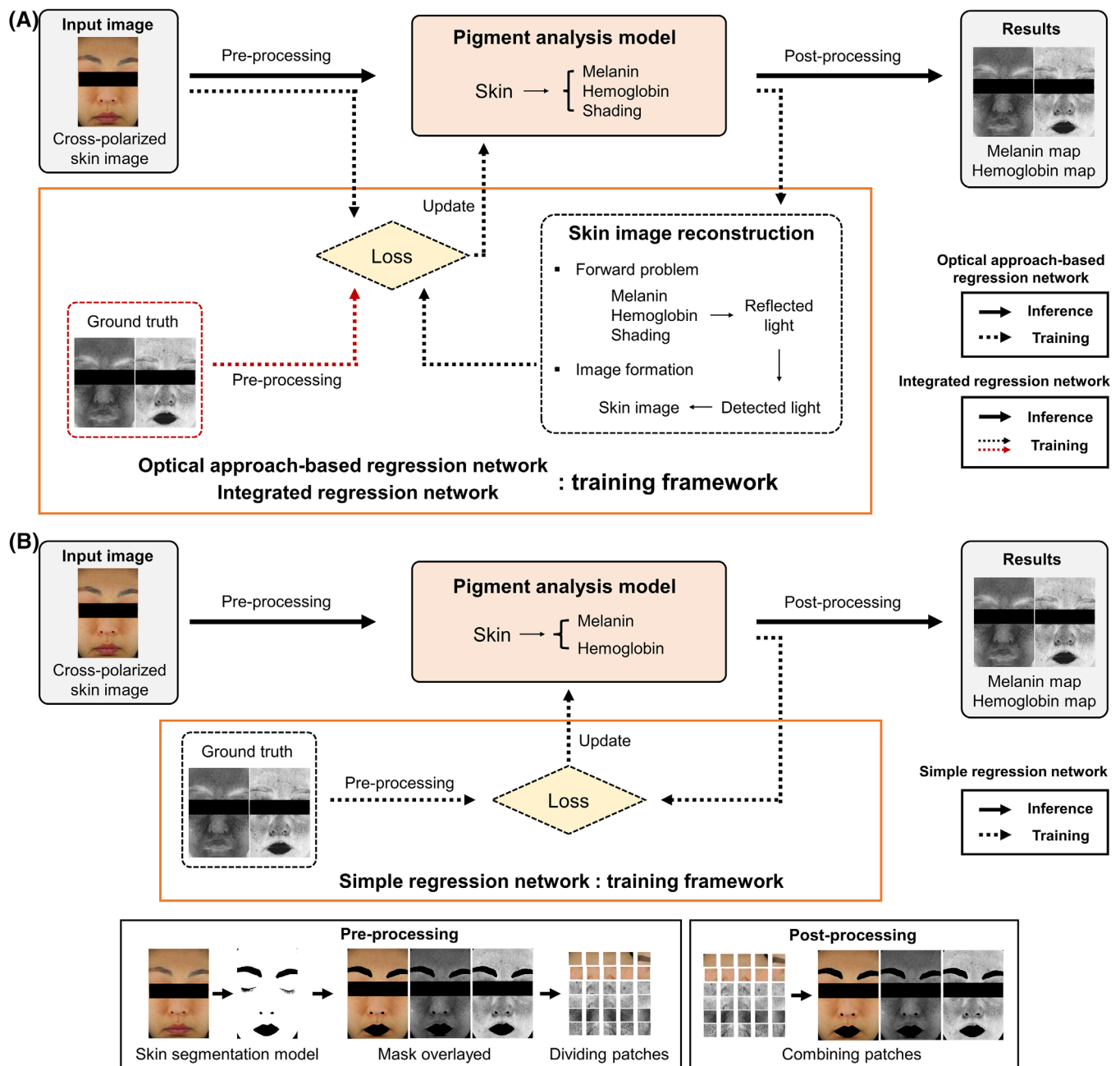


FIGURE 1 Overview of the training and inference frameworks used in the pigment analysis model. During pre-processing, the skin region is segmented as a binary mask, overlaid onto the input image, and subsequently divided into multiple patches. The pigment analysis model takes a patch of a cross-polarized image as input and produces melanin and hemoglobin maps. The training frameworks for three regression networks are depicted: “simple regression network,” “optical approach-based regression network,” and “integrated regression network.” (A) In the training frameworks of both the optical approach-based regression network (the black dotted line) and the integrated regression network (both the black and red dotted lines), the pigment analysis model decomposes skin regions of patch images into melanin, hemoglobin, and shading maps. The output maps are reconstructed into the skin image by solving the forward problem, with training based on a loss function minimizing the discrepancy between the patch image and the reconstructed image. The training framework of the integrated regression network incorporates the ground truth for pigments within the learning structure of the optical approach-based regression network, and integrates the correlation between the ground truth and the output patches for melanin and hemoglobin into the loss function to simplify the learning process. During inference, the output patches derived from the trained network are combined to form melanin and hemoglobin maps for the entire skin region. (B) The simple regression network is trained to predict outputs resembling the prepared ground truth. To analyze higher resolution images that exceed the input size limitation, the input image is divided into multiple patches and combined, following the same approach as the integrated regression network.

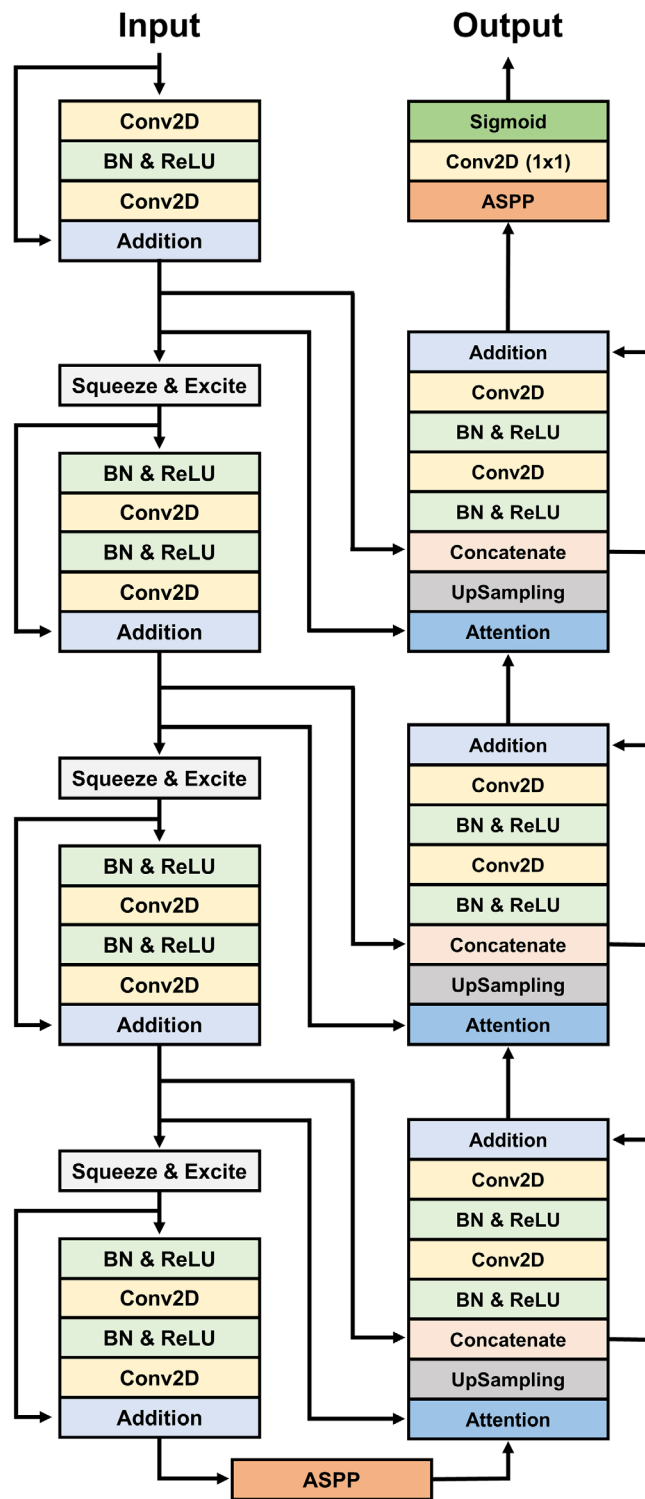


FIGURE 2 The architecture of the ResUNet++ model combines the ResNet and UNet structures. The model features a stem block and three sets of squeeze and excitation blocks (SE) and encoder blocks. The astrous spatial pyramid pooling (ASPP) module is located in the bottleneck, followed by three decoder blocks. Both the encoder and decoder blocks consist of two repetitions of batch normalization and ReLU activation (BN&ReLU) and convolution layers (Conv2D).

been instances where the loss value did not change significantly over several hundred epochs. To alleviate these limitations, we made modifications to the model architecture and the training framework, as depicted by the black dotted line in Figure 1A. We set the input type to ideally captured cross-polarized images, which eliminate the effect of specularly reflected light on the captured image, thereby excluding the specular map from the output. This change simplifies the learning process by reducing the number of outputs. We also employed a more advanced architecture to enhance performance. These modifications have resulted in improved performance and shorter training times, making the revised training framework and architecture more suitable for the optical approach-based regression network used in this study.

In this section, we describe the theoretical process of acquiring skin images. We analyze the optical properties of the primary factors influencing image formation from the perspective of the light spectrum. We term this methodology the “optical approach” in this study.

2.2.1 | Pre-processing and post-processing

The procedures involved in pre-processing and post-processing are demonstrated in Figure 1. In the pre-processing stage, the skin region is segmented as a binary mask and overlaid onto the input image. Then, the overlaid image is divided into multiple patches, referred to as the “patch image” in this study, that serve as input to the model. In the post-processing stage, the model's output patches are combined to create full-sized pigment maps of the same size as the original input image.

Training the pigment analysis model using the optical approach required us to exclude non-skin regions exhibiting distinct optical properties. For this purpose, we employed a skin segmentation model based on the ResUNet++ architecture. This model produced a binary mask that effectively masked non-skin regions such as background, hair, eyebrows, eyelashes, eyes, nostrils, and mouth. We trained the skin segmentation model on a dataset comprising 2100 images, combining both cross-polarized and normal skin images with a resolution of 640×480 pixels. The test set included 232 images of both types. The model successfully segmented two classes (skin and non-skin) and achieved a complete intersection over union (CIoU) [18] score of 0.961.

To overcome the input image resolution constraints imposed by computing resources, we adopted a patch-based strategy. This strategy involves dividing the input image into several patches and combining the output patches, assuming consistent lighting and camera conditions across

the entire image. Each patch includes overlapping areas with adjacent ones to manage the issue of visible boundary lines between patches. In the overlapping areas, weighted blending is applied with opposite gradients. As the training process is performed on individual patches instead of the combined image, the training dataset does not include the overlapping area. For training purposes, images are divided into patches of size 310×310 pixels and are randomly cropped to 256×256 pixels for augmentation, which matches the model's input size. For inference, images are divided into patches of size 256×256 pixels, which include a 15-pixel overlapping area. To restore the linear characteristics of images for accurate analysis, the color space of the input patch image is converted from standard RGB (sRGB) to linear RGB using inverse gamma correction.

2.2.2 | Image formation

The inverse problem approach of determining the optical properties of skin from measured reflected light is highly ill-posed, due to the skin's diffusive property [19]. In contrast, we trained our model using a forward problem approach that derives the spectrum of the reflected light from the given pigment conditions, which are the model's outputs. This approach eliminates the need for iterative calculations as the forward problem has a unique solution. Furthermore, it does not require ground truth data for pigments and can be applied to various image

acquisition systems, provided that a corresponding system's training dataset is available.

The skin image formation process is based on a spectral approach, as shown in Figure 3. When the illuminated light encounters the boundary between the skin and air, part of the illuminated light undergoes specular reflection due to the difference in refractive index, while the remaining light refracts into the skin layers. Inside the skin, the refracted light is absorbed and scattered before being reflected back out. Both the specularly reflected light and diffusely reflected light, detected by the camera, contribute to skin image formation. Professional skin diagnostic equipment used for measuring pigment regions [8, 20] selectively captures the diffusely reflected light as a cross-polarized image, providing enhanced features of concentrated melanin and hemoglobin areas. To incorporate this into our pigment analysis model, we restricted the input type to cross-polarized images and minimized the influence of specularly reflected light from the image formation process and the output maps. This adaptation aims to simplify the learning process by reducing the number of outputs.

Based on the presence of chromophores, the skin layers are classified into the epidermis layer with absorption dominated by melanin and the dermis layer with absorption dominated by hemoglobin. In terms of scattering properties, the epidermis has a negligible effect due to its small thickness, while the dermis, composed of elastin and collagen fibers, significantly contributes to

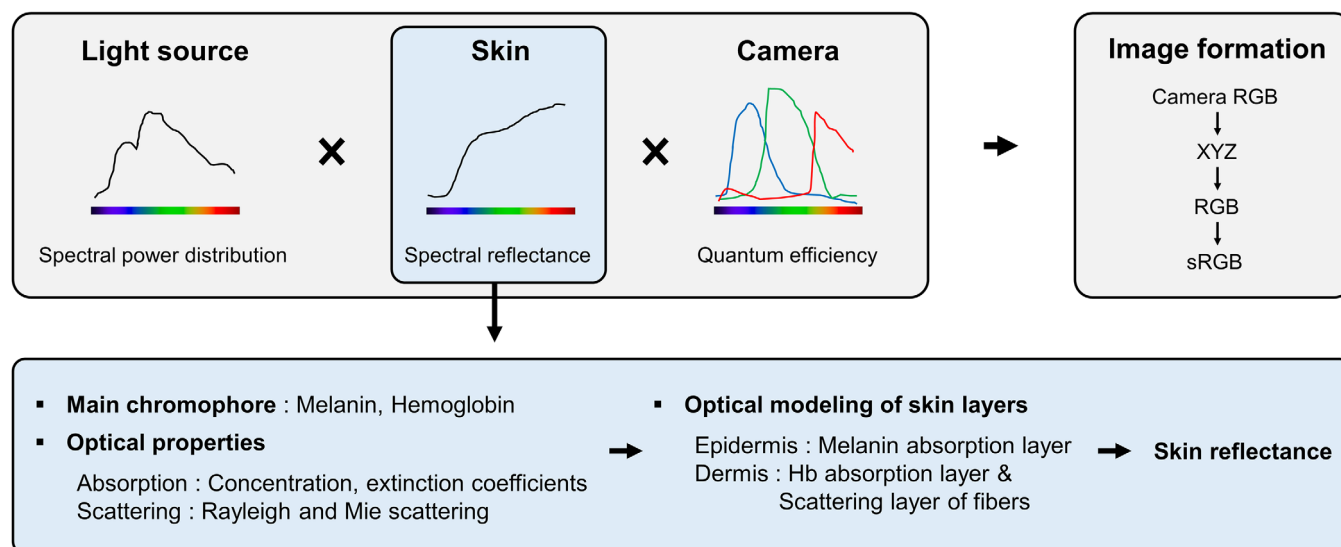


FIGURE 3 The skin image formation process initiates with incident light interacting with the skin. The reflected light captured by the camera sensor generates a skin image. This process involves the multiplication of the spectral power distribution of the light source, the skin's spectral reflectance, and the quantum efficiency of the camera sensor. The skin reflectance spectrum is approximated by optically modeling the skin layers, which factors in the absorption of chromophores and overall scattering. The resultant image is initially produced in a camera-specific RGB color space and is subsequently converted to the sRGB color space.

scattering and is considered the scattering-dominant layer [5].

The absorption and reduced scattering coefficients are defined by the following equations [21]:

$$\mu_a(\lambda) = \sum_i V_i \epsilon_i(\lambda) C_i \quad (1)$$

$$\mu'_s(\lambda) = a \left(f_{\text{Ray}} \left(\frac{\lambda}{\lambda_0} \right)^{-4} + \left(1 - f_{\text{Ray}} \right) \left(\frac{\lambda}{\lambda_0} \right)^{-b_{\text{Mie}}} \right) \quad (2)$$

Here, the absorption coefficient $\mu_a(\lambda)$ is calculated as the sum of the product of the volume fraction V_i , the extinction coefficient ϵ_i , and the concentration C_i of the chromophores i in melanin and hemoglobin. The scattering component comprises the Rayleigh and Mie scattering terms, which depend on the scattering amplitude a , the scattering power b , and the fraction of the Rayleigh scattering events f_{Ray} . In this study, the wavelength λ is defined as the visible range from $\lambda_{\min} = 450$ nm to $\lambda_{\max} = 750$ nm.

The skin reflectance is determined using a simplified form of the Kubelka-Munk theory, which assumes that the majority of diffusely reflected light originates from the dermis layer. The equation is as follows [22]:

$$R(\lambda) = T_{\text{epidermis}}(\lambda)^2 R_{\text{dermis}}(\lambda) \quad (3)$$

Here, the skin reflectance $R(\lambda)$ consists of the fraction of the transmitted light through the epidermis layer $T_{\text{epidermis}}(\lambda)$ and that of reflected light from the dermis layer $R_{\text{dermis}}(\lambda)$. Previous studies [17, 23] have utilized skin reflectance data derived using the Kubelka-Munk theory, considering the volume fractions of melanin (ranging from 1.3% to 43%) and hemoglobin (from 2% to 7%) [24, 25] along with their corresponding absorption spectra [26]. In this study, we adopt the same skin reflectance data for accurate analysis.

The skin image is formed and converted using the following equations:

$$I_m = \int_{\lambda_{\min}}^{\lambda_{\max}} L(\lambda) (R(\lambda) M_{\text{shading}}) C_m d\lambda \quad (4)$$

$$I_{\text{rgb}} = M_{\text{xyz to rgb}} \times M_{\text{cam to xyz}} \times (M_{\text{wb}} \times I_{\text{cam}}) \quad (5)$$

Here, the generated image I_m is obtained by integrating the spectral power distribution (SPD) of the light source $L(\lambda)$, the skin reflectance $R(\lambda)$, the shading map M_{shading} , and the quantum efficiency of the camera sensor C_m over the wavelength range from $\lambda_{\min} = 450$ nm to

$\lambda_{\max} = 750$ nm for each channel m in r , g , and b . The resulting image $I_{\text{cam}} = [I_r, I_g, I_b]$ is initially in the device-dependent color space. It is then sequentially converted to the XYZ color space, which is a standardized color representation [27], and further transformed to the RGB color space using the multiplication of the corresponding conversion matrices M . Furthermore, when saving the inferred reconstructed image, we convert the image's color space from RGB to sRGB using a gamma correction.

2.2.3 | Loss function

The model is trained to minimize the discrepancy between the patch image and the reconstructed image, which is derived from the output maps. Under identical acquisition conditions (using the same camera and light source), a higher similarity between the input and reconstructed image suggests that the model has successfully learned to create output maps which closely resemble the pigment distributions of the input image. The loss function used is the mean squared error (MSE) during training and the peak signal-to-noise ratio (PSNR) during the evaluation process.

2.3 | Integrated regression network

2.3.1 | Training method

The optical approach-based regression method offers analysis results comparable to those from professional skin diagnostic equipment, which is demonstrated in Section 3 and our previous research [15]. Despite the capabilities of this approach, it does have certain limitations. Throughout the training process, the loss function evaluates only the reconstructed image. This image is influenced by all outputs, including melanin, hemoglobin, and shading maps, making it difficult to verify the accurate training of individual outputs. For example, incorrect combinations of outputs can possibly generate colors resembling the patch image. Additionally, evaluations emphasizing representative values for the entire patch image could overlook the localized inaccuracies.

To overcome these limitations, we introduce an integrated regression method. This technique combines the learning structure of the optical approach with ground truth data for pigments, as shown in Figure 1A. While the optical approach's primary strength lies in its ability to train without the need for ground truth, the integrated regression method incorporates ground truth into the training process to enhance performance. Differing from simple regression method, this method aims to support to

correctly generate pigment maps while maintaining the skin image formation process. As a result, it enhances performance while retaining the benefits of the optical approach, such as combining output patches without occurrence of boundary lines and the generation of pigment-modified images.

The data intended for ground truth was generated using the VISIA system [28], a diagnostic tool prevalent in the industry, which we explain in more detail in Section 2.4. This system produces pigment concentration maps, making it compatible with our model. In our previous study [15], we noted that the VISIA system demonstrated a similar trend, but presents more accurate results.

2.3.2 | Loss function

The loss function is defined as follows:

$$L_{\text{total}} = L_{\text{recon}} + \alpha L_{\text{pigment}} \quad (6)$$

$$L_{\text{recon}} = \frac{1}{n} \sum_{i=1}^n (I_{\text{input}(i)} - I_{\text{recon}(i)})^2 \quad (7)$$

$$L_{\text{pigment}} = \sum_{k=\text{mel}, \text{Hb}} \left(\frac{\sum_{i=1}^n (M_{k(i)} - \overline{M}_k) (GT_{k(i)} - \overline{GT}_k)}{\sqrt{\sum_{i=1}^n (M_{k(i)} - \overline{M}_k)^2 \sum_{i=1}^n (GT_{k(i)} - \overline{GT}_k)^2}} \right) \quad (8)$$

Here, the total loss function L_{total} is computed as the weighted sum of the reconstruction loss L_{recon} and the pigment loss L_{pigment} . The reconstruction loss measures the mean squared error between the input image I_{input} and the reconstructed image I_{recon} for i all pixels n . The pigment loss is the summation of the Pearson correlation coefficient for the melanin and hemoglobin components, which involve the output maps M_k , the corresponding ground truth GT_k , and their respective mean values for all pixels $\overline{M}_k$ and $\overline{GT}_k$.

While determining the weight value α , iterative attempts should be made, taking into account intermediate results and dataset-specific adjustments. A high weight value might produce results similar to the regression network, potentially creating issues like differing scales in different patches or visible boundary lines. In this study, we started with the weight value α of 0.00005 for the first 150 epochs, adjusting it to 0.001 for the remaining epochs.

2.4 | Experiments

We received the dataset from the Global Medical Research Center using the VISIA-CR acquisition system

(Canfield Scientific, Inc). A total of 88 subjects participated in the data acquisition process, with all procedures complying with national regulations and institutional policies governing research ethics. Prior informed consent was obtained from all participants, and the study received approval from the authors' institutional review board. The VISIA VAESTRO image analysis system's RBX-Brown and RBX-Red Processing 1.0 modules [28] were utilized to generate melanin and hemoglobin distribution maps. This system is widely recognized as a reliable diagnostic tool for both research and clinical applications [20, 29].

Our pigment analysis model was trained on a dataset consisting of cross-polarized images with a resolution of 3648×5472 pixels. The training set comprised 77 subjects (1540 patches), and the test set included 11 subjects (220 patches). We did not separate a validation set in order to efficiently utilize the limited dataset for training, and we will address it further in the discussion (Section 3). The captured image (3648×5472 pixels) was cropped and resized to 1240×1550 pixels, subsequently divided into 20 patches (4×5) of 310×310 pixels. During training, these patches were randomly cropped to 256×256 pixels, and data augmentation techniques such as horizontal flipping and rotation were applied using the "Albumentations" library [30]. The test dataset was cropped and resized to 979×1461 pixels, and divided into 20 (4×5) patches, each 256×256 pixels in size. Each patch incorporated a 15-pixel overlap to maintain the resolution of the original image (979×1461) when combining the patch results.

We evaluated the performance of pigment analysis model trained with three different frameworks: the simple regression network, the optical approach-based regression network, and the integrated regression network. These networks were trained using the same dataset under the image acquisition conditions of the standard illuminant "D65" and the "Canon 5D Mark II" camera model [31]. The results were then compared with those from the VISIA system. Evaluation metrics included correlation coefficients [32] for the melanin and hemoglobin maps, and PSNR and the structural similarity index (SSIM) between the reconstructed and patch images. Especially for the integrated regression network, PSNR and SSIM metrics were indicative of the skin image formation process's preservation after incorporating ground truth into the optical approach-based learning structure. We also conducted a qualitative comparison in pigmented regions to identify potential discrepancies not captured by numerical analysis. From our observations, integrating ground truth into the optical approach requires extra stabilization time due to the introduction of extra constraints in the loss function. For more efficient training, we initialized the integrated regression

network using weights from the optical approach-based regression network, which was trained for 250 epochs on our training dataset.

Our models were trained on the PyTorch platform using an RTX 3090 GPU. We implemented a batch size of 16, a learning rate of 0.002, and the Adam optimizer with a cosine annealing scheduler. The total training time to reach 900 epochs was as follows: 11.5 h for the simple regression network, 17 h for the optical approach-based regression network, and 13.5 h for the integrated regression network.

3 | RESULTS

The performance of the three regression networks—the simple regression network, the optical approach-based regression network, and the integrated regression network—is compared to that of the VISIA system, as illustrated in Figure 4. The networks' performances are evaluated using correlation coefficients, PSNR, and SSIM, computed with the test dataset consisting of 11 subjects.

Melanin and hemoglobin maps obtained from the VISIA system, which are colored brown and red respectively, are converted to grayscale for direct comparison. To modify the pigment regions, either to enhance or alleviate them, the melanin and hemoglobin maps are multiplied by a constant. Similarly, the overall skin tone is adjusted by adding a constant to the pigment maps.

The performance metrics of the three regression networks are presented in Table 1. For reference, we have also included the simple regression network using a full-sized image as input. This table reports the correlation coefficients for the hemoglobin and melanin maps, as well as the PSNR and SSIM for the reconstructed image.

4 | DISCUSSION

Figure 4 compares the performance of the three regression networks to the VISIA system. Figure 4A shows the change in correlation coefficients of melanin and hemoglobin over the training epochs. It is observed that all regression networks stabilize their correlation coefficients after about 300 epochs. The optical approach-based regression network exhibits relatively lower correlation coefficient values and a wider standard deviation than the other networks, particularly in melanin analysis. The simple regression network and the optical approach based regression network yield similar coefficients, but the integrated regression network presents a wider standard deviation in hemoglobin analysis. To evaluate the similarity between the input and reconstructed images,

which indirectly reflects pigment analysis capability during training, Figure 4B presents the changes in PSNR and SSIM. These metrics validate whether the skin image formation process is maintained when integrating the ground truth during training. The simple regression network is excluded from this comparison as its learning structure does not encompass image formation.

Given that the incorporation of ground truth into the optical approach-based learning structure requires substantial time to stabilize during training, we applied pre-trained parameters for the integrated regression network, resulting in higher initial performance compared to the model trained solely with the optical approach. Even after ground truth integration, the PSNR values consistently exceed 30 dB. Such PSNR values suggest that the reconstructed image closely resembles the input image, making them difficult to distinguish by the human eye [33, 34], which confirms the effective preservation of the image formation process.

Table 1 presents a comparison of the regression networks' performance, including the simple regression network with a full-sized input image as a reference. In the simple regression network, the network using a full-sized image exhibits a higher correlation coefficient for hemoglobin than the network using a patch image, due to the absence of patch combination issues that can arise when combining patches. Nonetheless, training and inferring on full-sized images demand significant memory capacity and computational resources, which restricts the maximum image resolution. The optical approach-based regression network mitigates this limitation by using multiple patch images, although its correlation coefficient for melanin is lower compared to the simple regression network. This might be due to the challenge of differentiating between skin darkness and shading, both of which exhibit similar variations to melanin. However, by referencing the ground truth, the correlation coefficient of melanin increases by 7.9% to 0.978 and hemoglobin by 2.3% to 0.975, with both seeing a decrease in standard deviation. This approach demonstrates slightly higher correlation coefficients than the simple regression network (full) but also presents wider standard deviations, indicating a trade-off between accuracy and consistency.

Figure 4C presents the melanin and hemoglobin maps generated for the same input image. The overall tendencies of the entire image appear to be similar, but noticeable differences are observed in the enlarged images presented in Figure 5. First, Figure 5A shows the comparison of the simple regression network and the integrated regression network. The simple regression network generally exhibits consistent patch results, with a few disparate regions appearing after merging.

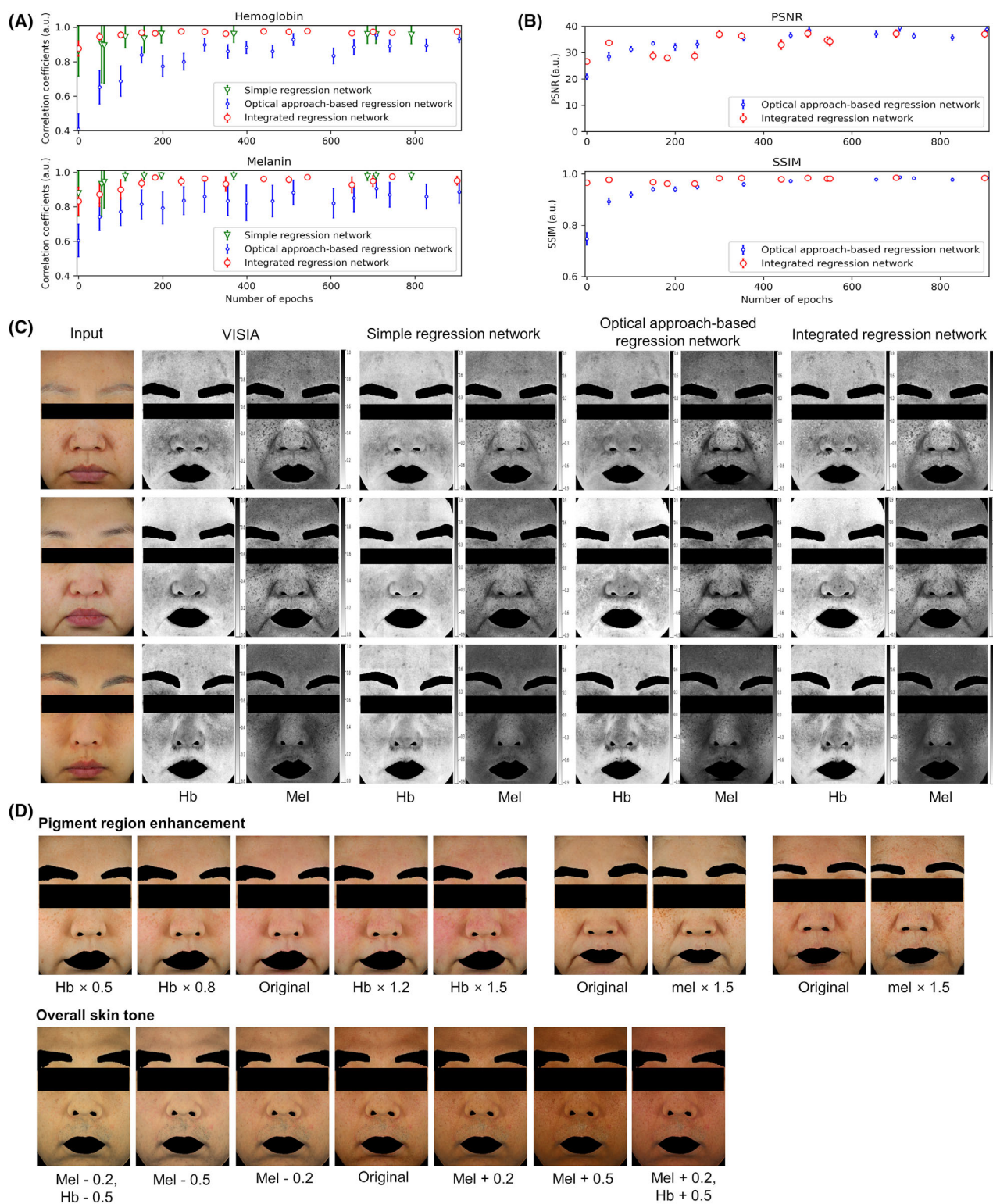
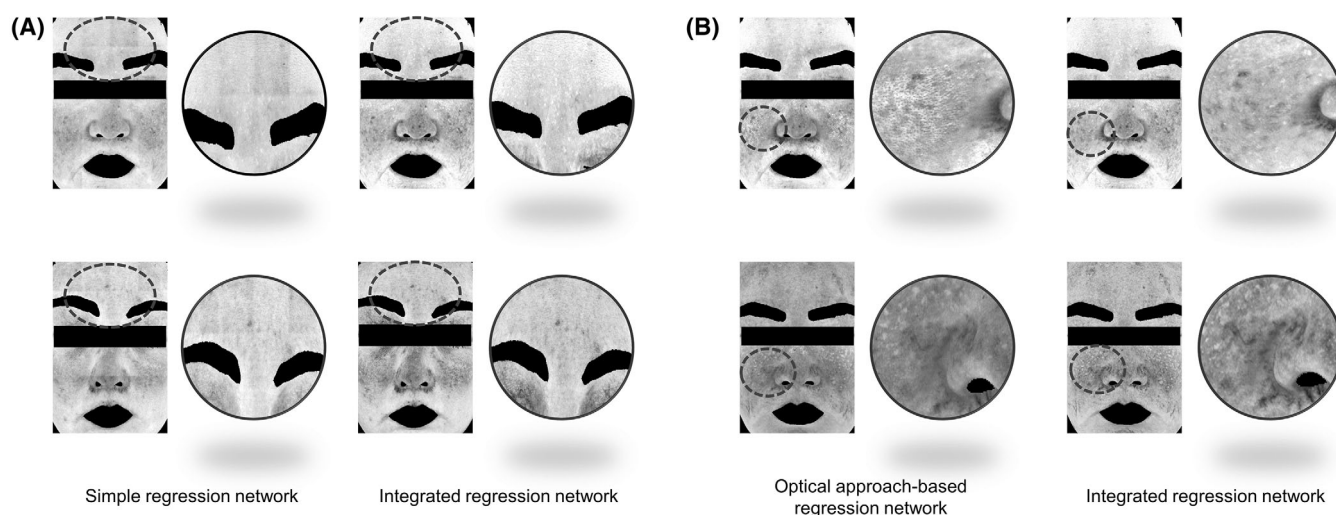


FIGURE 4 Comparison of the pigment analysis model trained with three different frameworks against the VISIA system: the simple regression network, the optical approach-based regression network, and the integrated regression network. (A) Correlation coefficients between the three networks and the VISIA system for (upper) hemoglobin and (lower) melanin maps. (B) (upper) PSNR and (lower) SSIM between the reconstructed image and the input image. (C) Comparison of the resulting hemoglobin and melanin maps. The first column presents the input image, followed by output maps from the VISIA system and the three networks in subsequent columns. (D) Reconstructed images produced by altering the pigment maps from the integrated regression network. The upper row demonstrates separate adjustments to the hemoglobin and melanin regions, maintaining the overall skin tone, while the lower row illustrates modifications to the overall skin tone.

TABLE 1 Performance of pigment analysis model trained with three different frameworks: the simple regression network (combining multiple patches vs. full-sized image), optical approach-based network, and integrated regression network.

	Hb (corr.)	Mel (corr.)	Recon. (PSNR)	Recon. (SSIM)
Simple regression (patches)	0.958 ± 0.052	0.977 ± 0.022	-	-
Simple regression (full)	0.971 ± 0.008	0.976 ± 0.005	-	-
Optical approach-based	0.953 ± 0.057	0.906 ± 0.125	39.411 ± 1.305	0.987 ± 0.003
Integrated regression	0.975 ± 0.027	0.978 ± 0.025	34.713 ± 1.330	0.981 ± 0.004

**FIGURE 5** Comparison of enlarged hemoglobin maps generated from the three regression networks. (A) Comparison of the simple regression network and the integrated regression network (B) Comparison of the optical approach-based regression network and the integrated regression network, focusing on the same regions of the same subjects.

Occasional variations in patch scale and boundary lines are observed. In contrast, the integrated regression network consistently generates the same results, regardless of whether it processes local parts or the entire image at once, a uniformity attributable to the application of consistent light source and camera conditions across all pixels.

Figure 5B demonstrates the changes when incorporating ground truth into the optical approach-based learning structure. The optical approach-based regression network may produce local regions with saturated pixel values below 0 or above 255 in some cases. Incorporation of ground truth during training adjusts the overall scale of the map, repositioning the previously saturated values within the valid image range and recovering accurate information, as depicted in the upper images. Furthermore, the previously blurred areas become clearer, and the contrast is enhanced in the lower images. The observed change suggests that the optical approach-based regression network generates intermediate values for ambiguous pixels, likely to prevent an increase in the loss function, with the ground truth providing clear guidance for these pixels.

In addition to its ability to analyze high-resolution images, the integrated regression network offers the advantage of generating modified skin images by manipulating the pigment maps during the image formation process, as shown in Figure 4D. By multiplying the pigment maps by a constant, the values of the pigment regions undergo significant changes, while the rest of the image remains relatively unchanged. As a result, the severity of the pigment regions can be modulated while preserving the overall skin tone in the upper images. Adding a constant to the pigment maps modifies the overall skin tone, as demonstrated in the lower images. These pigment-modified images highlight the versatility of the optical approach, which can simulate post-treatment images and create pigment-augmented datasets for pigment analysis or segmentation models.

In this study, the data were divided into training and test datasets without validation dataset, primarily due to two reasons. The first reason involves the discrepancy between our evaluation criteria during training and after training. During training, our focus is centered on the

network's image reconstruction capability, whereas post-training evaluations compare the derived pigment maps with the VISIA system. The second reason is due to practical limitations in obtaining datasets of cross-polarized images and analyzed results. Therefore, rather than allocating resources to create a separate validation set, we increased the ratio of the training dataset. Undoubtedly, if the quantity of the dataset is sufficient, establishing a validation set would be optimal.

A significant advantage of the proposed approach is its versatility. While most skin analysis tools are dependent on the image acquisition equipment of a specific company, the same training framework and model architecture in our study can be applied to various types of image acquisition equipment. The proposed approach can be further enhanced by considering several strategies. Firstly, a larger dataset could improve performance by exposing the model to a wider variety of melanin and hemoglobin combinations, thus enhancing its accuracy. In this study, there was an additional benefit of dividing the images into multiple patches, which increased the size of the training dataset by a factor of 20. Additionally, incorporating ground truth for the shading map using image processing techniques could potentially enhance the training process, similar to the way we incorporated ground truth for pigments. From an application perspective, integrating an additional model to convert the colors from any image acquisition conditions to those of the training dataset would significantly increase the versatility of the proposed approach. Lastly, calibrating the scale values using a precise contact-type clinical device could yield more accurate and reliable measurements. These future directions hold great promise for advancing the proposed approach and expanding its applicability in various domains.

5 | CONCLUSION

In this study, we introduced an integrated regression method that combines the optical approach with ground truth for pigment analysis. Our model overcomes the limitations of the model's input size by dividing images into multiple patches, enabling detailed analysis. The incorporation of ground truth improves the network's performance, leading to more accurate pigment analysis results. Furthermore, the versatility of the integrated regression network is demonstrated through its ability to generate pigment-modified images for various potential applications. We believe that with further advancements and refinements, our proposed approach will find wider applicability in dermatology and beyond.

AUTHOR CONTRIBUTIONS

Geunho Jung was involved in writing the manuscript and implemented a skin analysis model. Semin Kim made technical contributions in implementing deep learning models. Jongha Lee and Sangwook Yoo were involved in preparing datasets and project management. All authors contributed to reviewing the manuscript. Geunho Jung is the corresponding author of this paper.

ACKNOWLEDGMENTS

This research was supported by National IT Industry Promotion Agency (NIPA) for computing resources. The research dataset was supplied by Global Medical Research Center (GMRC, Republic of Korea), and the IRB approval number is "GIRB-21426-PD".

CONFLICT OF INTEREST STATEMENT

The authors have no relevant financial interests in this article and no potential conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

ORCID

Geunho Jung  <https://orcid.org/0000-0002-4896-009X>

REFERENCES

- [1] G. Zonios, A. Dimou, I. Bassukas, D. Galaris, A. Tsolakidis, E. Kaxiras, *J. Biomed. Opt.* **2008**, *13*, 014017.
- [2] P. Agache, P. Humbert, *Measuring the Skin*, Springer Science and Business Media, Berlin and Heinderberg, **2004**.
- [3] I. Saknite, S. Kwun, K. Zhang, A. Hood, F. Chen, L. Kangas, P. Kortteisto, A. Kukkonen, J. Spigulis, E. R. Tkaczyk, *J. Biophotonics* **2023**, *16*, e202300009.
- [4] G. N. Stamatias, B. Z. Zmudzka, N. Kollias, J. Z. Beer, *Pigm. Cell Res.* **2004**, *17*, 618.
- [5] L. E. Dolotov, Y. P. Sinichkin, V. V. Tuchin, S. R. Utz, G. B. Altshuler, I. V. Yaroslavsky, *Lasers Surg. Med.* **2004**, *34*, 127.
- [6] A. Treesirichod, S. Chansakulporn, P. Wattanapan, *Indian J. Dermatol.* **2014**, *59*, 339.
- [7] G. Zonios, J. Bykowski, N. Kollias, *J. Invest. Dermatol.* **2001**, *117*, 1452.
- [8] Y. Zuo, A. Li, H. He, R. Wan, Y. Li, L. Li, *Skin Res. Technol.* **2022**, *28*, 846.
- [9] Q. Marin, T. Honda, H. Nakajima, Y. Okano, M. Cherel, E. Prestat-Marquis, *Skin Res. Technol.* **2022**, *28*, 582.
- [10] N. Tsumura, H. Haneishi, Y. Miyake, *Opt. Rev.* **2000**, *7*, 479.
- [11] S.-H. Tseng, P. Bargo, A. Durkin, N. Kollias, *Opt. Express* **2009**, *17*, 14599.
- [12] H. M. Son, W. Jeon, J. Kim, C. Y. Heo, H. J. Yoon, J.-U. Park, T.-M. Chung, *Sci. Rep.* **2021**, *11*, 5350.

- [13] Qian Zheng, Ankur Purwar, Heng Zhao, Guang Liang Lim, Ling Li, Debasish Behera, Qian Wang, Min Tan, Rizhao Cai, Jennifer Werner, Dennis Sng, Maurice van Steensel, Weisi Lin, Alex C. Kot, arXiv preprint arXiv:2203.16056 **2022**.
- [14] M. Kim, M. H. Song, *Appl. Sci.* **2023**, 13, 989.
- [15] G. Jung, S. Kim, J. Lee, S. Yoo, *J. Biomed. Opt.* **2023**, 28, 035001.
- [16] D. Jha, P. H. Smedsrud, M. A. Riegler, D. Johansen, T. De Lange, P. Halvorsen, H. D. Johansen, *2019 IEEE International Symposium on Multimedia (ISM)*, IEEE, San Diego, CA, **2019**, p. 225.
- [17] Sarah Alotaibi, William Smith, arXiv preprint arXiv: 1908.10578. **2019**.
- [18] X. Chang, J. Wang, W. Yang, Y. Lei, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, IEEE Computer Society, Los Alamitos, CA, **2021**, p. 1192.
- [19] J. Feng, Q. Sun, Z. Li, Z. Sun, K. Jia, *J. Biomed. Opt.* **2019**, 24, 051407.
- [20] X. Wang, X. Shu, Z. Li, W. Huo, L. Zou, Y. Tang, L. Li, *Skin Res. Technol.* **2018**, 24, 379.
- [21] Lisa Kobayashi Frisk **2016**.
- [22] M. Doi, S. Tominaga, *Color Imaging VIII: Processing, Hard-copy, and Applications*, SPIE, Bellingham, WA, **2003**, p. 221.
- [23] A. Krishnaswamy, G. V. G. Baranoski, *Computer Graphics Forum*, Vol. 23, Wiley Online Library, Hoboken, NJ, **2004**, p. 331.
- [24] M. A. Ansari, R. Massudi, *Opt. Lasers Eng.* **2009**, 47, 965.
- [25] S. L. Jacques, R. D. Glickman, J. A. Schwartz, *Laser-Tissue Interaction VII*, SPIE, Bellingham, WA, **1996**, p. 468.
- [26] SL Jacques, *Skin Optics*. Oregon Medical Laser Center News, **1998**.
- [27] S. Bianco, F. Gasparini, A. Russo, R. Schettini, *IEEE Trans. Consum. Electron.* **2007**, 53, 1020.
- [28] R. Demirli, P. Otto, R. Viswanathan, S. Patwardhan, J. Larkey, *Canfield Systems White Paper* **2007**, 1, 1.
- [29] A. Goldsberry, C. W. Hanke, K. E. Hanke, *J. Drugs Dermatol JDD* **2014**, 13, 1312.
- [30] A. Buslaev, V. I. Iglovikov, E. Khvedchenya, A. Parinov, M. Druzhinin, A. A. Kalinin, *Information* **2020**, 11, 2.
- [31] J. Jiang, D. Liu, G. Jinwei, S. Süssstrunk, *IEEE Workshop on Applications of Computer Vision (WACV)*, Vol. 2013, IEEE, Los Alamitos, CA, **2013**, p. 168.
- [32] M.-T. Puth, M. Neuhäuser, G. D. Ruxton, *Anim. Behav.* **2014**, 93, 183.
- [33] X. Zeng, L. Ping, Z. Li, *J. Multimedia* **2009**, 4, 3.
- [34] C.-C. Lee, W. Hsien-Chu, C.-S. Tsai, Y.-P. Chu, *Pattern Recognit.* **2008**, 41, 2097.

How to cite this article: G. Jung, S. Kim, J. Lee, S. Yoo, *J. Biophotonics* **2023**, 16(12), e202300231.
<https://doi.org/10.1002/jbio.202300231>